MILITARY SPECIFICATION

TRUCK, MAINTENANCE; POWER LINE INSULATOR WASHER, 34,000 POUNDS GVW, DED OR GED, 4 BY 2

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers a diesel or gasoline engine driven, four-wheel, two rear-wheel drive truck having a gross vehicle weight (gvw) of not less than 34,000 pounds (lb), equipped with a power line insulator washer system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and Standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified (see 6.1), the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the soliciation.

SPECIFICATION

FEDERAL

W-B-131 - Battery, Storage: Vehicular, Ignition, Lighting and Starting.

|Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.|

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STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-281 - Automobiles, Trucks, Truck Tractors and Dollies; Preservation and Packaging of.
MIL-STD-1223 - Administrative Wheeled Vehicles Treatment, Painting, Rustproofing, Undercoating, Identification Marking, Data Plates and Warranty Notice Standards.
MS51317 - Light, Warning, Vehicular, Rotating, D.C.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines: Certification and Test Procedures.
Interstate Motor Carrier Noise Emission Standards.
Motor Vehicle Air Pollution Standards.

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852; or CFR, Title 40 should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

STATE OF CALIFORNIA

California Vehicle Code.

(Application for copies should be addressed to the Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

(Copies of specifications, standards and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117 - Standard Method of Salt Spray (Fog) Testing.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Description. Unless otherwise specified (see 6.1), the truck shall be a diesel engine driven, four-wheel, two rear-wheel drive chassis with cab having a gvw rating of not less than 34,000 lb. The truck shall be equipped with a power line insulator washer system consisting of a hydraulic operated boom with turret, remote controlled spray nozzle, water tank, and all required equipment as specified herein.

3.2 First production vehicle. The contractor shall furnish a truck for first production vehicle inspection.

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3.3 Standard commercial product. The truck, components, and accessories shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the truck, components, and accessories being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

* 3.3.1 Drain plugs. Drain plugs installed in components for the purpose of draining lubricants and hydraulic oils shall be of the permanent magnet type.

3.3.2 Heavy-duty cooling system. A heavy-duty cooling system shall maintain engine coolant at a temperature below the boiling point with truck loaded to rated gvw and operated at an altitude of 10,000 feet above sea level, or in an ambient air temperature of not less than 125 degrees Fahrenheit (°F).

3.3.3 Coolant system and indicators. Coolant system shall include a de-aeration system, a surge tank or a coolant recovery reservoir of not less than 2-quart capacity. A low coolant level or high coolant temperature alarm buzzer and red indicator warning light shall be provided on the cab dash instrument panel.

3.3.4 Towing devices. Not less than two hooks or loops for towing the truck shall be furnished on the front of the truck.

3.3.5 Splash guards and shields. Rear wheels shall have rigid splash shields ahead of the wheels and splash guards to the rear. Splash and stone throw protection shall be in accordance with SAE J682.

* 3.3.6 Engine hour meter. An engine hour meter, having a totalizing mechanism of not less than 9,999 hours, shall be furnished for the truck chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous trouble-free performance under severe operating conditions. Engine hour meter shall be mounted on the cab instrument panel or in a readable location under the hood.

3.3.7 Air pollution control. Trucks shall comply with the EPA regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, truck(s) destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

3.3.8 Sound level. The interior sound level shall conform to DoT Federal Motor Carrier Safety Regulation, section 393.94. The vehicle exterior sound level shall conform to EPA Noise Emission Standards for Transportation Equipment, Medium and Heavy Trucks.

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3.3.9 Radio noise suppressors. The truck shall be equipped with radio noise suppressors conforming to SAE J552. Electromagnetic radiation from the truck shall be within the limits of SAE J551.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Truck weights, ratings, and dimensions.

3.5.1 Curb weight. Curb weight shall include weight of chassis and cab, with all attachments, accessories, equipment, body, full complement of fuel, lubricant, and coolant.

3.5.2 Weight rating. The gvw rating shall consist of curb weight, operator and passenger (weight computed at 175 lb each), and payload, to provide not less than the specified gvw.

3.5.3 Weight distribution. The distribution of the gvw for the purpose of establishing suspension, axle, and tire capacities shall be determined with the payload uniformly distributed over the load area. The gross axle weight rating shall be not less than 12,000 lb on the front and 22,000 lb on the rear.

3.5.4 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. Minimum gvw shall be 34,000 lb.

3.6 Performance.

3.6.1 Speeds and gradeability. High speed requirements shall be met with truck loaded to specified gvw.

3.6.1.1 High speed gradeability. The truck shall ascend a continuous grade of 1.0 percent at 50 miles per hour (mph). Gradeability requirement shall be met with the main transmission in direct drive. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

3.6.1.2 Maximum geared speed. Maximum geared speed at engine governed speed shall be not less than 55 mph. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Engine governed speed}}{\text{Total geared reduction X tire factor}}$$

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3.6.2 Brake performance. Service brakes shall comply with the performance requirements specified in DoT Federal Motor Carrier Safety Regulations, section 393.52. Service brakes shall control and hold the truck, loaded to the rated gvw, on a 30 percent grade. Parking brake shall hold the truck, loaded to the rated gvw, headed either up or down the grade, without slipping on a 30 percent grade.

3.7 Chassis components.

* 3.7.1 Engine. The engine furnished for the specified vehicle class shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

* 3.7.1.1 Diesel engine. Unless otherwise specified (see 3.4.1.2), the vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine, with not less than four cylinders on class A and not less than six cylinders on classes B and C. Engine net horsepower figures used in performance prediction calculations shall be determined in accordance with SAE J816 or SAE J1349. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling.

* 3.7.1.2 Gasoline engine. When specified (see 6.1), the vehicle shall be equipped with a liquid cooled, internal combustion, four-stroke cycle gasoline engine with not less than six cylinders. The engine furnished shall produce the required vehicle performance when operated on unleaded fuel with a research octane rating of 91, at engine speed not more than manufacturer's recommended operating speed. The engine shall be capable of warranted operation on unleaded fuel when used in accordance with the operator's manual. Unless otherwise specified or required by foreign laws, vehicles for overseas used shall be capable of accepting and operating on leaded gasoline. The rated net torque output of the engine for class A trucks shall be not less than 265 pound-feet. Class B and class C trucks and class A and class B truck tractor rated net engine torque shall be not less than 295 pound-feet. Class C truck tractor rated net engine torque shall be not less than 320 pound-feet. Engine net torque ratings and net horsepower figures used in performance preduction calculations shall be determined in accordance with SAE J245 or SAE J1349. A fan clutch shall be provided. The fan clutch shall reduce fan speed automatically when the fan is not required for engine cooling.

* 3.7.1.3 Oil filter. A full flow type oil filter shall be furnished. For civil agencies, an independently mounted minimum 10-quart oil filter with replaceable element shall be furnished for diesel engine driven vehicles.

* 3.7.1.4 Governor. An engine governor shall be furnished and set and sealed to limit the engine to the engine manufactuer's recommended operating speed.

* 3.7.1.5 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. On diesel engine driven vehicles, the control shall include partial thermostatic control of coolant flow through the radiator and thermostatically controlled radiator shutters or complete thermostatic control of all coolant flow through the radiator.

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* 3.7.2 Electrical system. The electrical system shall be in accordance with DoT Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.7.2.1 Starting system. For diesel engine driven vehicles, a 12- or 24-volt (V) starting system, with a 12V direct current (dc) lighting system shall be furnished. A minimum 60-ampere (A) alternator shall be furnished. Engine starting equipment shall include an ether starting system or a glow plug. If an ether system is furnished in lieu of a glow plug, it shall be of the measured shot type. Measured shot type ether systems shall be key operated or manually operated from the driver's compartment, and shall be inoperative with the engine warm. Complete provisions for a replaceable ether reservoir of not less than 12 fluid ounces shall be furnished. A reservoir need not be furnished.

* 3.7.2.2 Ignition system. For gasoline engine driven vehicles, a 12V dc ignition system shall be furnished. An alternator of not less than 40A rated capacity, and which provides not less than 16A dc output at normal engine idle speed, shall be furnished. When an electric motor driven hydraulic tailgate is furnished (see 3.1.1.12), or when a converter type 24V system is furnished (see 3.4.2.8), an alternator of not less than 60A rated capacity, and which provides not less than 16A dc output at normal engine idle speed, shall be furnished.

3.7.2.3 Lighting. All vehicle lights, reflectors, and wiring shall be as specified herein and shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.12, 393.19, 393.20 and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on vertical surface of rub rails unless recessed and fully protected or mounted on vehicle bumpers.

3.7.2.4 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. Vehicle shall be provided with double-faced front signal units and single-faced rear signal units installed in accordance with SAE J588. Turn signal operating units shall have visible and audible flash indicator.

3.7.2.5 Batteries. Each battery shall be of 12V potential. The total reserve capacity rating and the total cold cranking rating at 0°F, both measured in accordance with SAE J537, shall be not less than specified in table I. The batteries shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131.

TABLE I. Batteries.

Engine type	Reserve capacity (minutes)	Cold cranking (amperes)
Diesel (less than 220 gross hp)	320	1200
Diesel (220 gross hp and over)	320	1800
Gasoline	100	450

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3.7.3 Fuel system. Fuel system shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.65 and 393.67.

3.7.3.1 Air cleaner. An air cleaner shall be furnished. When equipped with an oil-bath type, oil capacity shall be not less than 1 quart.

* 3.7.3.2 Fuel tank(s). Fuel tank(s) shall be not less than 50 gallons total capacity (43 gallons for vehicles destined for California). When more than one fuel tank is furnished on diesel engine driven vehicles, means shall be provided to assure equalized fuel level in both tanks. When more than one tank is furnished on gasoline engine driven vehicles, a selector valve connecting either tank to engine fuel intake shall be provided and means shall be provided to monitor the fuel level of either tank from a single fuel gage.

3.7.4 Exhaust system. The exhaust system shall conform to DoT Federal Motor Carrier Safety Regulation, section 393.83. Exhaust mufflers, when stacked, shall be furnished with a heat shield. Stacked exhaust systems shall be provided with a hinged rain cap.

3.7.5 Automatic or semiautomatic transmission. The vehicle shall be provided with an automatic or semiautomatic transmission. The transmission furnished shall provide continuous drive. The transmission shall include a hydraulic torque converter and not less than five forward gear ratios. Normal driving range selector position shall provide not less than four gear ratios without movement of the selector.

3.7.6 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.7.7 Frame. The chassis frame shall be the manufacturer's standard. Truck shall have a heavy duty main frame or frame reinforcements. The heavy duty frame shall have frame rails of greater section modulus than the manufacturer's standard for the vehicle furnished and shall provide structural strength at least equivalent to the reinforced frame specified herein for the type vehicle furnished. Reinforcements shall extend at least from the rear of the front suspension, rear hanger bracket to the front of the rear spring, front hanger bracket. Frame reinforcements shall be in accordance with the body manufacturer's recommendations. Frame rails shall not project beyond the rear end of the body.

3.7.8 Suspension. Truck shall be equipped with suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with truck loaded to rated gvw. When suspension capacity is rated at the spring pads, unsprung weight shall be deducted. Truck shall be equipped with hydraulic, double-acting, shock absorbers at the front wheels.

3.7.9 Axles. Axle ratings shall be at least equal to that specified in 3.5.3.

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3.7.9.1 Traction control. The vehicle shall be furnished with a traction control on the rear axle. The traction control shall actuate automatically to insure that power is transmitted to the wheels having traction when the opposite wheel losses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

3.7.10 Wheels, rims, tires, and tubes. Truck shall be equipped with single front and dual rear wheels. Rim and tire ratings shall conform to TRA recommendations, for the type and size of tires furnished. Tire and rim sizes shall be the same for all wheels on each truck. Wheels shall be of the 10 stud disc type.

3.7.10.1 Tires. Tires shall be tube or tubeless bias ply type with highway tread. Tires shall be of rated capactiy at least equal to the load imposed on each tire, measured at each wheel, at the ground, with truck loaded to rated gvw.

3.7.10.2 Tubes. Tubes, when furnished, shall be of heavy-duty type, and shall be of proper size for tires furnished. Tire flaps shall be provided in accordance with TRA recommendations.

3.7.10.3 Spare wheel. A spare wheel shall be furnished and stowed on the truck.

3.7.10.4 Tire chain clearance. Tire chain clearance in accordance with SAE J683 shall be provided. Allowance for spring deflection shall be included.

3.7.11 Brakes. Brakes shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.40 through 393.43 and 393.45 through 393.52.

3.7.11.1 Service brakes. The vehicle shall be equipped with full-air brakes. The braking system, complete with all necessary components, shall include:

a. Air compressor, unloader-head-type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less that 12 cubic feet per minute.
b. Air storage reservoir(s) with not less than 2,000 cubic inches total capacity, each tank equipped with drain, and with safety and check valves between compressor and last reservoir tank.
c. Foot control, suspended or treadle type.
d. Air control valves.
e. Air pressure gage, visible to the driver.
f. Low-air-pressure warning, visible and audible.
g. Service brake stop lamp switch.
h. Alcohol aspirator with unbreakable transparent container.
i. Automatic moisture ejector.

3.7.12 Cab. The manufacturer shall furnish any standard or optional, full width, conventional model cab. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, and with at least one door equipped with external, key-operated lock. Drip rails shall be installed above the cab doors. Cab shall have upholstered, full-width, adjustable seat

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and back or individual adjustable driver's seat and individual passenger seat. The color of the upholstery and the interior finish shall be compatible with the exterior color. White upholstery shall not be furnished. Safety grips shall be provided on each side of the cab to assist personnel to enter or leave the cab. Interior dome lighting shall be provided. Manufacturer's standard fastenings and two pairs of seat belts shall be furnished and installed on the truck.

3.7.13 Steering. Power steering shall be furnished.

3.7.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type and operated by either air or electric motor(s).

3.7.15 Bumper. Unless the bumper is an integral part of vehicle cab, a channel type bumper shall be provided.

3.7.17 Tool stowage. Stowage space of sufficient size to accommodate vehicle jack, hand tools, antiskid chains, and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. Stowage space for these tools may be furnished inside the cab. When stowage space for these tools is located outside of cab, it shall be waterproof and shall provide for locking with a padlock.

3.7.18 Heater and defroster. Hot water heater shall be provided. Heater shall have fresh air intakes. Discharge outlets shall be provided to direct air to floor and to defroster louvers. Heater shall be complete with blower and mounted controls convenient to the driver.

3.7.19 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of operator to and from driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Instruments shall be panel mounted.

3.7.20 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

a. Key-operated ignition switch.
b. Ammeter, charging indicator, or voltmeter.
c. Fuel gage (each tank).
d. Oil pressure gage or indicator.
e. Speedometer with recording odometer.
f. Engine temperature gage or indicator.
g. Dual sunvisors.
h. Front door mounted armrest on driver's side.
i. Driver's compartment ventilator other than window.
j. Manufacturer's standard electric horn.

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3.7.21 Rearview mirror. Outside rearview mirrors shall be mounted on each side of the cab. The mirrors shall be of the combination type having flat and convex areas inclosed in a common housing. The flat portion shall have not less than 50 square inches of reflective area. The convex portion shall have not less than 20 square inches of reflective area. On tilt cabs, the mirrors shall have not less than two supports.

3.7.21.1 Operation mirror. A flat surface mirror shall be furnished for driver observation of operator. The mirror shall have not less than 65 square inches of flat reflective area. The mirror shall be mounted to the left rearview mirror bracket in such a manner to not obstruct normal driving vision yet give driver full mirror view of operator.

3.7.22 Back-up alarm. The truck shall be provided with an audible, pulsating, signaling device (electrical or mechanical) to caution personnel when the vehicle is in reverse gear operation.

3.8 Body. The truck shall be equipped with a power line insulator washer system. The unit shall include a hydraulic operated boom, turret, water tank, auxiliary engine, water and hydraulic pumps, and all required equipment specifed herein. The boom turret, water tank, auxiliary engine, and outriggers shall be mounted on suitable subframes. The subframes shall not be welded to the truck chassis.

3.8.1 Boom. A hydraulically operated telescoping or articulator type boom shall be furnished. The boom, when fully extended, in its most vertical position shall have a tip height of not less than 60 feet above ground level. The boom shall be mounted on a turret with capability of continuous 360° rotation without stops. The boom and turrent shall operate from the auxiliary engine powered hydraulic pump. The boom and turret shall be capable of withstanding the total thrusts created by washer spray nozzle at any boom angle and any boom extension. The boom, when fully elevated, shall attain an angle of not less than 85°. The retracted length shall be not greater than 27 feet measured from the centerline of the base to tip. A boom extension lock to positively lock the boom in the extended position shall be furnished.

3.8.1.1 Boom angle indicator and stability chart. A boom angle indicator shall be furnished and mounted readily visible to the seated operator. A vehicle stability chart shall be furnished to indicate stability of chassis for all positions, angles and extensions of boom without utilizing outriggers, and also when utilizing the outriggers.

3.8.1.2 Boom storage. For normal storage, travel, and shipment, the boom shall lower to a horizontal position along the centerline of the vehicle. A supporting frame shall be furnished with a saddle type rest to support the boom. The supporting frame shall be mounted to the truck frame.

3.8.2 Boom tip wash nozzle. A hydraulic controlled wash nozzle shall be mounted to the tip of the boom. The wash nozzle mounting shall have hydraulic actuators which provide for a minimum of 90° left and right horizontal movement and a minimum of 60° above and below level with boom at any angle of elevation. The nozzle movement shall be controlled by a single lever (joy stick) located on operator's control panel.

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3.8.3 Boom hose retractor assembly. When a telescopic boom is furnished a hose retracting device shall be provided for the water and hydraulic hoses which extend to the boom tip. The device shall be mounted on the street side of the boom.

3.8.4 Outriggers. A-frame, telescoping type outriggers shall be mounted to the rear of the turret base section. The outriggers shall be hydraulically operated with controls located at rear of platform. A second set of controls shall be mounted in the truck cab for driver operation. The outriggers, when fully extended, shall have a spread dimension of not less than 13-feet measured to the outside of ground pads. Each ground pad shall have not less than 100 square inches of flat surface. A safety system shall be provided which shall prevent the vehicle being moved except with outriggers in fully retracted position.

3.8.5 Water tank. The water tank shall be a rectangular low profile type with a capacity of not less than 1,000 gallons. The tank shall be fabricated of mild steel and shall have an interior finish coating of epoxy. The tank shall be baffled. A 16-inch access/filler hole with lockable hinged cover shall be furnished on top of tank. The bottom of tank shall be furnished with proper sized threaded openings for installation of pump suction, pressure relief return, water level detection probe, and water quality detection electronic probe. Plumbing for external fill and drain shall be furnished. The tank shall have subframe for mounting to truck chassis.

3.8.5.1 Water tank level monitor. An electronic water level monitor consisting of a tank probe and indicator panel shall be furnished. The indicator panel shall be of the five lite type indicating water levels in one quarter tank increments and a tank refill indicator. The water level monitor shall be connected to allow water warning system which activates a warning horn at the one quarter level and activates engine shutdown to idle at six-inch water level to prevent operation of pump without water.

* 3.8.6 Auxiliary engine. An auxiliary engine shall be furnished to power the water and hydraulic pump systems. The engine shall be a liquid cooled, internal combustion two or four-stroke diesel with not less than four cylinders. The engine shall provide the required power to operate the water and hydraulic system.

* 3.8.6.1 Engine components. The engine shall be furnished with at least the following manufacturer's standard components:

a. Electric starter, 12V.
b. Air cleaner (oil bath or dry type).
c. Engine hour meter (9,999 total hours).
d. Full flow oil filter.
e. Muffler with spark arrestor and rain cap.
f. Oil pressure gage with low pressure engine shutdown.
g. Water temperature gage with high temperature engine shutdown.
h. Ammeter or voltmeter.
i. Fuel tank with 8 hour capacity.

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3.8.6.2 Engine electrical system. The auxiliary engine electric system shall be 12V potential. The alternator shall have an output of not less than 35A. The battery shall have a total reserve capacity of not less than 100 minutes and total not less than 350 cold cranking ampere at 0°F. The battery shall be mounted inside a weatherproof container and located for ease of maintenance. The electrical system shall furnish all electrical requirements for the power line insulator washer systems.

* 3.8.6.3 Engine shutdown system. An electrical engine shutdown system shall be furnished to return engine to idle and divert water back to tank when:

a. Water resistance is at unsafe low.
b. Water tank level is below safe operating level.
c. Pump pressure is at an unsafe low.

A manual override of shutdown system shall be provided for operation of a hydraulic system to retract boom and outrigger.

3.8.6.4 Fuel system. The auxiliary engine shall draw its fuel from a separate tank. The auxiliary tank shall have fuel capacity for not less than 8 hours continuous operation.

3.8.7 Hydraulic system. The hydraulic fluid nonconductive system shall consist of the components required to perform all hydraulic operations of the boom assembly, and outriggers with the degree of control specified herein. Mechanism and reservoirs shall be positioned or shielded to prevent hazardous fluid contact with heated surfaces or electrical apparatus. An accumulator may be used as a component of the hydraulic system to provide the maximum degree of pressure control.

3.8.7.1 Hydraulic cylinders. All hydraulic cylinders shall incorporate seals to prevent entrance of moisture and dirt into the cylinders and leaks of hydraulic fluid from the cylinders. O-type packings shall be used in the cylinders where static seals are required. Manufacturer's standard seals shall be furnished where applied to reciprocating parts. All hydraulic cylinders for actuating the booms, and outriggers shall be provided with pilot retraction of the outriggers in case of loss of hydraulic pressure. Means shall be provided to install pressure gages for testing the hydraulic system pressure and relief settings. All surfaces of hydraulic ram piston rods shall be chrome plated with sufficient thickness to withstand a 30-hour salt spray test in accordance with ASTM B117.

3.8.7.2 Hydraulic lines and fittings. Hydraulic lines shall be located to facilitate maximum ease of removal and installation, or visual inspection for failures and maximum protection from damage during operation of the boom assembly. Flexible, hydraulic hose or collector rings shall be used where flexing of hydraulic lines is encountered and where installation of components requires a flexible coupling. Standard straight-thread, flared type fittings or adaptions to flared type fittings, or tapered pipe thread with dry seal fits shall be furnished. Hydraulic lines and fittings shall be air and liquid tight and shall be installed to preclude vibration and chafing. The system shall be designed to preclude hydraulic transients. Provisions shall be made for bleeding air from the hydraulic system and completely draining all

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hydraulic lines, when necessary. Hose assemblies and all seamless steel hydraulic lines shall be adequate to provide the rate of flow, temperature and pressures required to operate the system smoothly and efficiently. Galvanized pipe and copper tubing shall not be acceptable.

3.8.7.3 Hydraulic pump. A hydraulic pump with necessary pressure and flow capacities to operate all hydraulic components shall be provided. The hydraulic pump shall be driven by the auxiliary engine. The pumping system shall include an adjustable pressure relief valve or unloading valve with range of adjustments for all required operations.

3.8.7.4 Hydraulic fluid reservoir. A hydraulic fluid reservoir shall be provided and shall have sufficient capacity to hold the required volume of hydraulic fluid, with not less than 10 percent for expansion space. The reservoir shall be provided with a filler opening and removable oil strainer and vented filler cap, accessibly located to facilitate filling the tank from a one-gallon container using a funnel. The vented filler cap shall have means to preclude fluid from splashing out of the tank during movement of the vehicle. The filler cap shall be attached to the tank by a corrosion resistant chain to prevent loss. Provision shall be made for inspection and cleaning of the interior, and shall allow recovery of the fluid in a container. The reservoir shall have a plate to indicate the type of hydraulic fluid furnished, and the maximum fluid level in the reservoir with the booms folded for travel position and the hydraulic lines and components full of fluid. A sight level or dip stick gage shall be furnished to observe the level of fluid. The tank shall be clean. A plate shall be secured to the hydraulic fluid reservoir stating "USE ONLY RECOMMENDED TYPE HYDRAULIC FLUID".

3.8.8 Water pump. A four stage, ultra-high pressure centrifugal fire fighting type pump shall be furnished. The pump shall be a Waterous Model CPK-4 or equal. The pump shall develop a minimum of 60 gallons per minute at not less than 1,000 pound-force per square inch (psi). The pump shall be driven from the auxiliary engine. A pressure relief valve shall be furnished. A pressure gage shall be furnished at the pump discharge.

3.8.9 Water quality monitor. An electronic water quality monitor system shall be furnished. The system shall consist of a tank probe and a sensing control. The system shall provide continuous liquid monitoring for conductivity/resistivity properties of the wash water. The monitor control shall provide for manually adjustable resistance level. The system shall provide for engine shutdown to idle, to prevent water spraying, when the resistivity becomes lower than the preset resistance level. The monitor system shall be powered from the auxiliary electrical system.

3.8.9.1 Water monitor control panel. A weather tight enclosure shall house the water resistivity and water tank level indicator systems. The enclosure shall be mounted at flywheel end of engine over the hydraulic and water pumps. The housing shall have hinged door with a clear plastic or glass panel.

3.8.10 Hand line hose reel. A hose reel shall be furnished on rear of platform to contain the handline hose and handline wash gun. The hose reel shall have an electric motor drive rewind. Electric control switch shall be

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mounted on reel frame. The hose reel shall be furnished with 150 feet of non-conductive, 1-inch diameter, high pressure water hose. The reel shall be connected to the high pressure water system in parallel with the boom spray nozzle.

* 3.8.11 Hand held wash gun. A hand operated wash gun shall be furnished and connected to the 1-inch high pressure hand line hose. The wash gun shall be system pressure pilot operated to assure ease of feathering and positive shutoff when released. The activator trigger shall require minimum pressure. The nozzle tips shall be interchangeable with the boom tip nozzle tips. The wash gun shall be furnished with a quick disconnect for removal from water hose.

3.8.12 Operator contol station. An operator control station shall be mounted to the right side of the turret. The control station shall have a fully upholstered seat located for ease of all controls operation and a direct line of sight of the boom tip spray nozzle at all boom angles. The control station shall have all controls for all functions of boom, boom tip nozzle, and engine throttle.

3.8.13 Rear platform. A full width platform shall be furnished at rear of vehicle for access to operator control station, servicing of water tank, hydraulic reservoir, water hose reel, grounding reel, and hydraulic boom. The platform shall be full width of the dual rear wheels and shall extend forward, on each side of boom turret assembly over the rear tires to within 6 inches of the rear edge of the water tank. The rear of the platform shall be so designed to provide for rear end protection specified in 3.8.17. The platform shall be fabricated of diamond plate steel. An access ladder shall be furnished at the curbside rear corner of the platform.

* 3.8.14 Grounding wire. An 85-foot number 2 AWG insulated cable shall be furnished. The cable shall be attached to the chassis on one end and shall have a grounding clamp at the other. There shall be a permanent storage compartment adjacent to the handling hose reel to store cable in when not in use.

3.8.15 Warning light. An amber rotating vehicle warning light conforming to MS51317-4 shall be furnished. The light shall be mounted with reinforcement on the boom cradle frame on the street side of boom cradle. Warning light shall be operated by means of a separate switch with red indicator light located on instrument panel in cab and marked to indicate function of switch.

3.8.16 Portable resistivity tester. A portable tester shall be furnished for testing the conductivity/resistivity of the wash water. The portable tester shall have an electronic measuring device calibrated to read both in ohms per inch cubed and ohms per centimeter cubed. The tester shall be furnished with a high impact protective carrying case.

3.8.17 Spare spray head nozzles. Two sets of spare spray head nozzles, one each 1/4-inch and one each 5/16-inch, shall be furnished for the hand washer gun and the boom tip spray head.

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3.9 Treatment and painting. The truck shall be treated and painted in accordance with MIL-STD-1223. Finish color shall be gloss yellow No. 13538 of FED-STD-595.

3.10 Marking and data plates. Marking, registration numbers, and data plates shall be in accordance with MIL-STD-1223, as specified for the Navy.

3.11 Rustproofing. The truck cab, chassis, and chassis underside shall be rustproofed in accordance with MIL-STD-1223.

3.12 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated prior to delivery with type of lubricant specified in the operator's manual and grade of lubricant recommended for ambient temperature at the delivery point. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

3.13 Servicing and adjusting. Prior to acceptance of the truck by the Government, the contractor shall service and adjust the truck for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

a. Inflation of all tires.
b. Adjustment of brakes.
c. Proper functioning of all lighting and electrical systems.
d. Wheel alinement.
e. Adjustment of engine to include tune-up.
f. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.
g. Cooling system filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

The truck shall be conspicuously tagged to identify the lubricants and their temperature range.

3.14 Workmanship.

3.14.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.14.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

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3.14.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.14.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First production vehicle inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. This inspection shall include the examination of 4.3 and the test of 4.4.1 through 4.4.8. The purpose of the inspection shall be to determine vehicle conformity

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with the requirements of the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its right under the provisions of the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4.1 through 4.4.8, and the packaging inspection of 4.5. This inspection shall be performed on the samples selected in accordance with 4.3.

4.4 Examination. Each truck shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests.

4.4.1 Vehicle weight. The first production vehicle shall be weighed to determine gross weight and distribution of gross weight on front and rear axles. The truck shall be weighed with water tank full of water, all specified equipment, driver and passenger, and payload (see 3.5.2) to ascertain that the axles, suspension, and tires furnished are of adequate capacity for the imposed loads.

4.4.2 Road test. The first production vehicle shall be road tested with payload by the contractor to assure thant the vehicle will operate in accordance with contractual requirements.

4.4.3 Boom operational tests. The boom shall be subjected to not less than 10 complete operational cycles for each function of extension and retraction, rotation, and elevation. Boom extended height, retracted height, retracted length, elevation angle, and turret rotation angle shall be verified to determine conformance to 3.8.1.

4.4.4 Washing operation test. The water pump shall be engaged and a simulated washing test conducted. Not less than one tankfull of water shall be discharged through the boom mounted nozzle. The output capacity and discharge pressure for the water pump specified in 3.8.8 shall be verified. The boom mounted nozzle shall be operated throughout its rotational and elevation ranges to determine conformance to 3.8.2.

4.4.5 General operational tests. An operational tests of five complete cycles for the outriggers to verify satisfactory performance shall be conducted. An operational test shall be conducted to verify satisfactory performance of hand held wash gun.

4.4.6 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the government to refuse acceptance of all vehicles under the contract until corrective action has been taken.

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4.4.7 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and be the last truck shipped on the contract. The first production vehicle shall be reconditioned, including replacement of abnormally worn parts and paint touch-up or repainting, prior to delivery to enable it to be accepted as a contract item. The contractor shall maintain the first production vehicle in a serviceable condition for the duration of the contract.

4.4.8 Production truck tests. The contractor's testing system shall, as a minimum, assure each truck is capable of meeting performance requirements contained herein. At the point of final acceptance the contractor shall make available to the Government, records acceptable to the Government indicating that the servicing and adjusting required by 3.13 has been accomplished.

4.5 Packaging inspection. The vehicle shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Vehicle processing. The equipment shall be preserved and packaged in accordance with the contractor's standard practice. When specified (see 6.1), equipment shall be preserved, packaged, and marked in accordance with the mobile requirements of MIL-STD-281 with the level of preservation and packaging as specified (see 6.1).

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When Government documents are other than specified (see 2.1.1).
c. When gasoline engine is required (see 3.1 and 3.7.1.2).
d. When preservation and packaging in accordance with MIL-STD-281 is required, and the level of preservation and packaging required (see 5.1).

6.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of DoD Federal Acquisition Regulations (FAR) Supplement 27.410-6, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DoD FAR 27.410-6 are not invoked, the data shall be delivered in accordance with the contract requirements.

* 6.3 Subject term (key word) listing.

Truck
Maintenance
Insulator washer
Diesel engine

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Hydraulic controlled
Telescopic
Control station

Custodian:

Navy - YD

Preparing Activity:

Navy - YD

(Project 2320-N267)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-29211A(YD)	TRUCK, MAINTENANCE; POWER LINE INSULATOR WASHER, 34,000 POUNDS GVW, DED OR GED, 4 BY 2

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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